Article

Predicting the Trainability of Variational Quantum Circuits: A Data-Driven Model for Barren Plateaus

Md Habibur Rahman [1]

[1] Department of AI Convergence Engineering, Gyeongsang National University (GNU), Jinju, South Korea; habib@gnu.ac.kr

Abstract

Barren plateaus are the main obstacle to training quantum machine-learning models. On a barren plateau, the cost-function gradient shrinks exponentially as the circuit grows, and training stalls. Today the only reliable test is to measure it. That means sampling many gradients for every candidate circuit, which is expensive. We ask a different question. Can a classical model predict trainability from the circuit architecture alone? The architecture here means qubit count, depth, ansatz type, entanglement pattern, entangler gate, and cost-observable locality. We test this directly. We use an exact statevector simulator to build a dataset of 20,000 random variational circuits. For each circuit we estimate the gradient variance $\mathrm{Var}[\partial C/\partial\theta]$ from 200 random parameter vectors, using the parameter-shift rule. This variance is the label. A gradient-boosted regressor then predicts $\log_{10}$ Var with $R^2 = 0.685$ on held-out architectures. A classifier separates barren from trainable circuits with an area under the ROC curve of 0.991. We also test transfer to larger circuits. We train only on small circuits and test on the two largest unseen sizes ($n = 11, 12$). The model still reaches $R^2 = 0.719$. Permutation importance then recovers established theory. Cost-observable locality and entanglement structure matter most. We make no quantum-advantage claim. This is classical machine learning about quantum circuits. It offers a cheap way to screen ansätze, and it shows which design choices cause untrainability.

Keywords: barren plateaus; variational quantum circuits; quantum machine learning; trainability; gradient variance; statevector simulation; gradient boosting

1. Introduction

Variational quantum algorithms (VQAs) are the leading way to use noisy intermediate-scale quantum (NISQ) hardware [1]. A VQA works in three steps. It encodes a problem into a parameterized quantum circuit, called an *ansatz*. It then measures a cost function $C(\theta) = \langle\psi(\theta)|O|\psi(\theta)\rangle$. Finally, a classical optimizer updates the parameters θ. This simple loop underlies variational quantum eigensolvers, quantum approximate optimization, and most quantum machine-learning (QML) classifiers [2–4].

One problem stops this loop from scaling. It is the *barren plateau* (BP). On a barren plateau, the variance of the cost gradient vanishes exponentially in the number of qubits [5, 6]. The loss landscape becomes almost flat everywhere. Gradient-based training then stalls. Several causes are now known. Global cost observables trigger barren plateaus [7]. So do very expressive or deep ansätze [8], poor initialization [9], and hardware noise [10]. The key point is simple. A barren plateau is mostly a property of the circuit *design*, not of one particular parameter setting.

Received:
Accepted:
Published:

The usual way to check a circuit is to measure it. One samples many random parameter vectors, evaluates the gradients, and estimates their variance. This is exactly the cost we want to avoid when screening a large design space. It also becomes hardest for many qubits, which is where barren plateaus matter most. This motivates our central question. *Can a classical model predict a variational circuit's trainability from its architecture alone, without sampling any gradients?*

We answer this question empirically. We treat trainability prediction as a supervised learning problem over circuit *specs*. The features are cheap architectural descriptors. Each one is a small integer or a category. The label is the gradient variance. We compute it once, offline, by exact statevector simulation. We then ask two things. Does a model trained on this data generalize to unseen architectures? And does it transfer from small circuits to larger ones? The second question matters more, because large circuits are the expensive case.

The contributions of this article are as follows:

- We formulate barren-plateau diagnosis as a supervised prediction task. The input is a circuit architecture, and the output is a trainability label. We treat it as regression ($\log_{10} \mathrm{Var}[\partial C/\partial\theta]$) and as binary classification (barren vs. trainable).
- We release a reproducible pipeline. It generates a labeled dataset of 20,000 random circuits with an exact, dependency-light statevector simulator. Seeds are fixed, and rows are streamed to disk.
- We show that a gradient-boosted model predicts trainability with $R^2 = 0.685$ on held-out architectures. The classifier separates barren circuits with AUC 0.991.
- We show transfer across scale. Trained only on $n_q \leq 10$ qubits, the model still attains $R^2 = 0.719$ on larger, unseen circuit sizes.
- We recover established theory through permutation feature importance. Cost locality and entanglement structure dominate, which gives the model an interpretable, physics-consistent basis.

We also want to be clear about scope. We make *no* claim of quantum advantage. The model is classical, and the data is classically simulated. The contribution is a data-driven *diagnostic* for a known quantum-computing bottleneck [11,12].

2. Background and Related Work

2.1. Variational Quantum Circuits

We use a hardware-efficient ansatz [13]. It repeats one simple pattern. Each layer applies a single-qubit rotation to every qubit. Each layer then applies a fixed pattern of two-qubit entangling gates. With n qubits and L layers the state is

$$|\psi(\theta)\rangle = \prod_{\ell=1}^{L} \left[U_{\mathrm{ent}} \prod_{q=1}^{n} R_{a_{\ell,q}}(\theta_{\ell,q}) \right] |0\rangle^{\otimes n}, \tag{1}$$

where R_a is a Pauli rotation about axis $a \in \{x, y, z\}$. The term U_{ent} is a layer of CZ or CX gates. These gates are arranged linearly, circularly, or all-to-all. The cost is the expectation of a Pauli-Z string, $C(\theta) = \langle\psi(\theta)|O|\psi(\theta)\rangle$. We use two choices. A *local* cost uses $O = Z_0$. A *global* cost uses $O = Z_0 Z_1 \cdots Z_{n-1}$.

2.2. Barren Plateaus

A circuit family has a barren plateau if $\mathrm{Var}_\theta[\partial C/\partial\theta_k]$ decays exponentially in n [5]. Cerezo et al. [7] proved a key result about the cost function. Global cost functions cause barren plateaus even at shallow depth. Local costs can stay trainable up to $O(\log n)$ depth. Cost locality is therefore a first-order design knob. Expressibility [8] and entanglement also correlate with flatter landscapes. Some structured architectures avoid the problem, and quantum convolutional networks are a proven example [14]. A recent review consolidates these results [6].

Most of this work is *analytical*. It derives variance scaling laws for one ansatz family at a time. Our work is *empirical* and complementary. We do not prove a bound for a single family. Instead, we learn a predictive map across a heterogeneous space of architectures.

2.3. The Parameter-Shift Rule

Some gates are generated by an operator with two distinct eigenvalues. For those gates, two circuit evaluations give the exact gradient [15,16]:

$$\frac{\partial C}{\partial \theta_k} = \tfrac{1}{2}\left[C(\theta + \tfrac{\pi}{2}e_k) - C(\theta - \tfrac{\pi}{2}e_k)\right]. \tag{2}$$

We use this rule for every gradient sample. Our labels are therefore exact, and they carry no finite-difference bias.

2.4. Trainability, Simulability, and Honest Framing

Recent work urges caution about advantage claims. Cerezo et al. [17] argue that the structure which removes barren plateaus often also makes a model classically simulable. Bowles et al. [12] show that out-of-the-box classical models often match or beat QML models on standard tasks. Schuld and Killoran [11] argue that the field should de-emphasize "beating classical."

We adopt this stance. The object we predict is a property of quantum circuits. The predictor and its value, however, are entirely classical. We are not aware of prior work that casts barren-plateau severity as a directly *predicted* regression or classification target from cheap architectural features, and that also tests extrapolation across qubit count.

3. Materials and Methods

Figure 1 summarizes the full pipeline. It runs from a sampled circuit spec to a predicted trainability value.

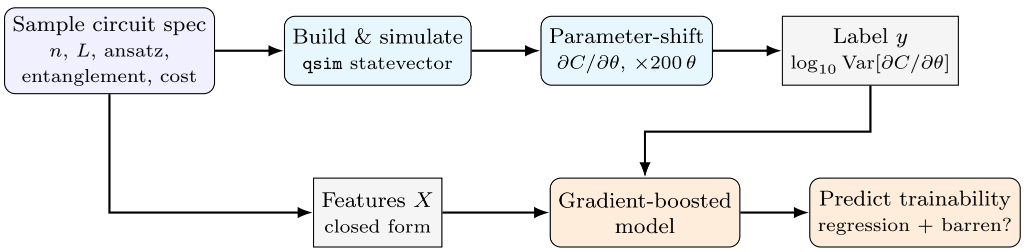

Figure 1. Method pipeline. A random circuit spec is built and simulated exactly with `qsim`. The parameter-shift gradient of the cost is sampled over 200 random parameter vectors to form the gradient-variance label y. Cheap closed-form architectural features X are read directly from the spec. A gradient-boosted model is trained on (X, y) to predict trainability.

3.1. Circuit Specification and Feature Space

Each datapoint is a *spec*. A spec fully determines one circuit family. Table 1 lists the feature vector we hand to the model. Every feature is computed in closed form from the spec. No quantum simulation is needed to obtain them.

Table 1. Architectural features used to predict trainability. All are derived in closed form from the circuit spec.

Feature	Meaning
`n_qubits`	number of qubits n
`n_layers`	number of ansatz layers L
`n_params`	$n \times L$ trainable rotations
`ansatz_type`	fixed-`ry` vs. random-Pauli axes
`entangle_pattern`	linear / circular / all-to-all
`entangler_gate`	CZ vs. CX
`cost_global`	local Z_0 (0) vs. global $Z^{\otimes n}$ (1)
`n_entanglers`	entangling gates $\times$ layers
`depth_ratio`	L/n

3.2. Label Generation

We label each spec in three steps. First, we draw 200 parameter vectors $\theta \sim \mathrm{Unif}[0, 2\pi)^{nL}$. Second, we compute the exact parameter-shift gradient of the cost. We always differentiate with respect to one fixed target rotation in the middle layer. Third, we record the empirical variance of those gradients.

The regression label is $y = \log_{10} \max(\mathrm{Var}[\partial C/\partial\theta], 10^{-18})$. The classification label is $\mathbb{1}[y < -2]$, which we call "barren." Statevectors are evolved exactly with the in-house `qsim` NumPy simulator [18]. The labels therefore describe ideal, noiseless trainability.

3.3. Dataset and Data Description

We sample 20,000 specs. Each design parameter is drawn independently and uniformly over the ranges in Table 2. Circuits use $n \in [2, 12]$ qubits and $L \in [1, 20]$ layers. Every spec is labeled with 200 gradient samples.

Generation is embarrassingly parallel. Every spec is seeded reproducibly from a root `SeedSequence`. Rows are streamed to CSV as they complete. The final dataset contains 46.7% barren circuits and 50.1% global-cost circuits. The classification target is therefore near-balanced, so a majority baseline is not competitive.

Table 2. Sampled design space. Each spec draws these parameters independently and uniformly.

Design parameter	Type	Sampled values
`n_qubits`	integer	2–12
`n_layers`	integer	1–20
`ansatz_type`	categorical (2)	fixed-`ry`, random-Pauli
`entangle_pattern`	categorical (3)	linear, circular, all-to-all
`entangler_gate`	categorical (2)	CZ, CX
`cost_global`	binary	local Z_0, global $Z^{\otimes n}$

Table 3 summarizes the numeric features and the regression label. The design spans small, easily simulated circuits at $n = 2$. It reaches $n = 12$, which is the boundary of exact statevector simulation here. Depth ranges from a single layer up to $L = 20$. The label spans roughly 18 orders of magnitude. It has a floor at -18, where the finite-sample variance underflows.

Table 3. Descriptive statistics of the numeric features and the trainability label over all 20,000 circuits.

Variable	Mean	Std	Min	25%	Median	Max
`n_qubits`	7.02	3.17	2.00	4.00	7.00	12.00
`n_layers`	10.48	5.80	1.00	5.00	11.00	20.00
`n_params`	73.56	55.55	2.00	28.00	60.00	240.00
`n_entanglers`	135.82	198.34	1.00	28.00	70.00	1320.00
`depth_ratio`	2.00	1.83	0.08	0.75	1.50	10.00
$\log_{10}$ Var (label)	-3.41	4.84	-18.00	-2.98	-1.85	-0.23

The labels reproduce established barren-plateau theory directly from the data (Table 4). Global cost observables give a mean $\log_{10}$ Var of -4.49, and local costs give -2.33. That gap is more than two orders of magnitude. Global costs also roughly double the barren fraction. This is exactly the cost-locality effect proved by Cerezo et al. [7]. Denser entanglement (all-to-all) and the CX entangler push circuits toward barren plateaus as well. This structure is what makes the prediction task learnable. It also gives the trained model an interpretable, physics-consistent basis.

Table 4. Trainability broken down by design choice. Lower mean $\log_{10}$ Var and higher barren % indicate less trainable families. The data recovers known barren-plateau drivers.

Design choice	Count	Mean $\log_{10}$ Var	Barren %
Cost-observable locality			
local Z_0	9974	-2.33	30.9
global $Z^{\otimes n}$	10026	-4.49	62.4
Entanglement pattern			
linear	6649	-2.50	38.5
circular	6748	-2.96	48.8
all-to-all	6603	-4.79	52.8
Entangler gate			
CZ	10096	-2.31	38.1
CX	9904	-4.53	55.5
Single-qubit rotation scheme			
fixed-ry	9878	-2.80	42.0
random-Pauli	10122	-4.01	51.3

3.4. Models and Evaluation

We do not want to favor a single learner. We therefore compare six model families. They span linear, instance-based, neural, and tree-ensemble inductive biases. The list is a linear baseline (ridge regression or logistic regression), k-nearest neighbours, a multilayer perceptron with two hidden layers (128–64), random forests, extremely randomized trees, and histogram-based gradient-boosted trees. We use the scikit-learn implementations [19]. Its histogram-based booster follows the design of LightGBM [20]. Features are standardized for the scale-sensitive models.

We report two protocols for each model. The first is a random hold-out split with 20% test data. It measures generalization to unseen architectures. The second is an *extrapolation* split. We train on $n \leq 10$ and test on all larger circuits. This is the headline test, because it asks whether small-circuit labels transfer to costlier regimes.

All models use fixed random seeds, so the reported numbers reproduce exactly. We assess interpretability with permutation feature importance on the best model. All code, command lines, and random seeds are released (see the Data Availability Statement). The underlying simulator is available at https://quantum.qbithabib.com.

4. Results

4.1. Model Comparison

Table 5 compares all six model families. It covers both tasks and both splits. Histogram-based gradient-boosted trees give the best held-out performance. They reach $R^2 = 0.685$ on regression and AUC 0.991 on classification. We therefore adopt this model for the rest of the paper.

Two observations sharpen the picture. First, the two tasks differ in difficulty. Classification is easy for every model, since all reach AUC ≥ 0.95. Regression is much harder, and the linear baseline falls far behind there ($R^2 = 0.27$ vs. 0.685). The architecture-to-trainability map is therefore nonlinear in its fine-grained structure. Simple rules of thumb do not capture it. Second, the neural network is competitive. It even edges out boosting on the extrapolation regression ($R^2 = 0.75$ vs. 0.72). This hints that a learned representation transfers slightly better to larger circuits. We still retain gradient boosting. It is stronger in-distribution, easier to interpret, and faster to train.

Table 5. Model comparison across both prediction tasks and both evaluation splits. Regression columns report R^2, classification columns report AUC; best hold-out result in **bold**.

	Regression R^2		Classification AUC	
Model	Hold-out	Extrap.	Hold-out	Extrap.
Hist Gradient Boosting	0.685	0.719	0.991	0.989
MLP	0.665	0.752	0.990	0.991
k-NN	0.645	0.639	0.986	0.979
Random Forest	0.640	0.716	0.985	0.988
Extra Trees	0.610	0.707	0.968	0.988
Linear baseline	0.274	0.297	0.953	0.889

4.2. Generalization to Unseen Architectures

We first use the random hold-out split. The regressor attains $R^2 = 0.685$ with mean absolute error 1.018 in $\log_{10}$ units. The classifier separates barren from trainable circuits with accuracy 0.950 and AUC 0.991. Figure 2 shows predicted versus true $\log_{10}$ gradient variance on the held-out set.

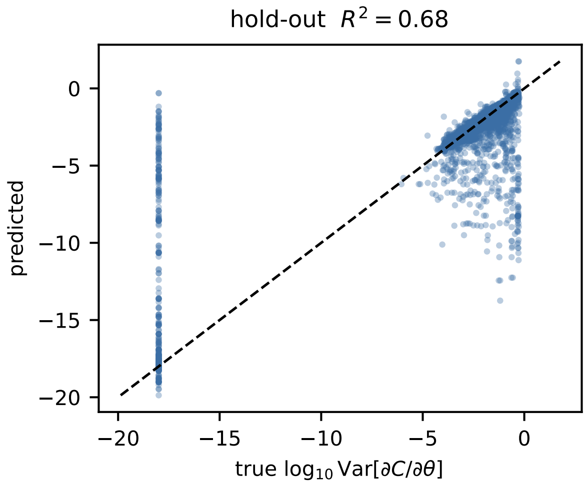

Figure 2. Predicted vs. true $\log_{10}$ Var$[\partial C/\partial\theta]$ on held-out architectures. The dashed line is $y = x$.

https://doi.org/10.3390/quantum1010000

4.3. Extrapolation across Qubit Count

We next test transfer to larger circuits. We train only on $n \leq 10$ qubits, which is 16 395 circuits. We test on the two largest unseen sizes, $n = 11$ and $n = 12$, which is 3 605 circuits. The regressor retains $R^2 = 0.719$ with MAE 1.264.

The extrapolation R^2 slightly exceeds the hold-out value. We do not read this as an easier task in general. The $n = 11, 12$ test set is dominated by saturated, strongly barren global-cost circuits. Their labels sit near the variance floor and are comparatively easy to predict. The result should therefore be read narrowly. Small-circuit labels transfer to the largest sizes we can still simulate. This is not evidence of unbounded extrapolation. Even so, the transfer is useful. The cost of ground-truth labeling grows exponentially with qubit count, so a predictor that moves downward in cost is what an ansatz-screening tool needs.

4.4. Trainability versus Qubit Count

Figure 3 shows the mean label as a function of qubit count. The curves are split by cost locality. Global-cost circuits show a steep, monotone decay of gradient variance with qubit number. That decay is the defining signature of a barren plateau. Local-cost circuits decay far more slowly. The dataset therefore captures the qubit-scaling behavior the predictor must learn.

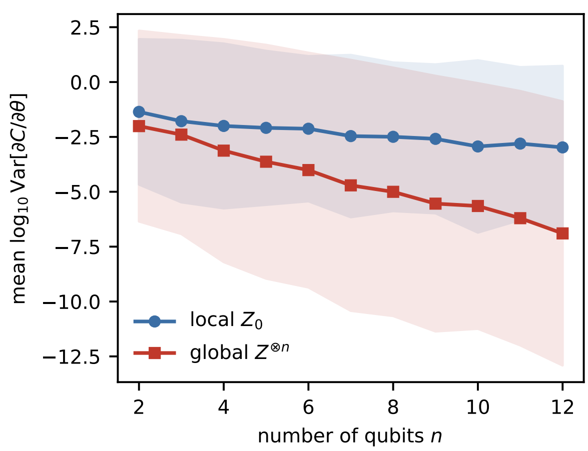

Figure 3. Mean $\log_{10}$ Var$[\partial C/\partial\theta]$ vs. qubit count for local and global cost observables (shaded: ± 1 standard deviation). Global costs show the exponential decay characteristic of barren plateaus.

4.5. What Drives Trainability

Figure 4 reports permutation feature importance. Cost-observable locality ranks first, and entanglement structure ranks next. Both agree with theory. Global costs [7] and denser entanglement [8] drive circuits toward barren plateaus. The model therefore rediscovers, from data alone, the design rules that the barren-plateau literature derived analytically.

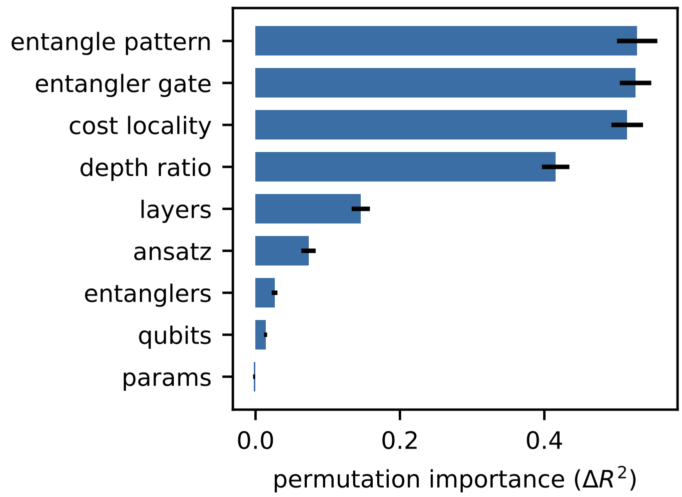

Figure 4. Permutation feature importance for the trainability regressor. Cost locality and entanglement structure dominate, matching barren-plateau theory [7,8].

5. Discussion

Our results support one main claim. Barren-plateau severity carries a strong architectural signature, and that signature is low-dimensional. A cheap classical model can recover it without any gradient sampling. The feature ranking is also consistent with physics, which raises confidence in the model.

This suggests a practical workflow. First, screen a large space of candidate ansätze with the learned model. Then reserve expensive gradient-variance estimation for the shortlist it flags as trainable.

Several limitations scope these claims. (i) Labels are limited to classically simulable qubit counts, so the extrapolation result is evidence of transfer, not a proof. (ii) We use an ideal statevector simulator, so noise-induced plateaus [10] are out of scope. (iii) The spec space is diverse, but it is still one hardware-efficient family. Other ansatz classes may need their own features. (iv) Gradient variance is estimated from finitely many samples, which adds label noise at the trainable end. (v) The label uses the gradient with respect to a single fixed middle-layer rotation. In principle the variance can depend on which parameter is probed. (vi) The barren cutoff ($\log_{10} \mathrm{Var} < -2$) is a modeling choice. Our primary regression target is threshold-free, so the cutoff affects only the binary framing.

Two design choices protect the results. Seeds are fixed and reported. The extrapolation split is defined purely by qubit count, so no large-circuit information leaks into training.

6. Conclusions

We recast barren-plateau diagnosis as supervised prediction. We then showed that a classical gradient-boosted model can predict a variational circuit's trainability from its architecture alone. The model is trained on statevector-simulated labels, and it transfers across qubit count. The approach offers a cheap ansatz-screening heuristic. It also gives a data-driven confirmation of which design choices cause untrainability.

Future work has three clear directions. The first is richer ansatz families. The second is graph or sequence encodings of circuits, paired with graph neural networks or transformers. The third is adding a noise model, so the predictor can also cover noise-induced plateaus.

Author Contributions: The author confirms sole responsibility for the following: study conception and design, software, data generation and curation, analysis and interpretation of results, visualiza-

tion, and manuscript preparation. The author has read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Acknowledgments: During the preparation of this manuscript, the author used ChatGPT (OpenAI) and Claude (Anthropic) for the purposes of language editing and grammar correction. The author has reviewed and edited the output and takes full responsibility for the content of this publication.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: All code (statevector simulator, dataset generator, training, and model-comparison scripts), the exact command lines, and random seeds are openly available at https://github.com/bithabib/quantum_computer_simulator. The dataset is regenerable end-to-end with a single command. Interactive demonstrations of the underlying simulator are available at https://quantum.qbithabib.com.

Conflicts of Interest: The author declares no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

BP	Barren plateau
VQA	Variational quantum algorithm
QML	Quantum machine learning
NISQ	Noisy intermediate-scale quantum
MAE	Mean absolute error
ROC	Receiver operating characteristic
AUC	Area under the ROC curve

References

1. Cerezo, M.; Arrasmith, A.; Babbush, R.; Benjamin, S.C.; Endo, S.; Fujii, K.; McClean, J.R.; Mitarai, K.; Yuan, X.; Cincio, L.; et al. Variational quantum algorithms. *Nature Reviews Physics* **2021**, *3*, 625–644. https://doi.org/10.1038/s42254-021-00348-9.
2. Havlíček, V.; Córcoles, A.D.; Temme, K.; Harrow, A.W.; Kandala, A.; Chow, J.M.; Gambetta, J.M. Supervised learning with quantum-enhanced feature spaces. *Nature* **2019**, *567*, 209–212. https://doi.org/10.1038/s41586-019-0980-2.
3. Farhi, E.; Neven, H. Classification with Quantum Neural Networks on Near Term Processors. *arXiv preprint arXiv:1802.06002* **2018**. https://doi.org/10.48550/arXiv.1802.06002.
4. Schuld, M.; Bocharov, A.; Svore, K.M.; Wiebe, N. Circuit-centric quantum classifiers. *Physical Review A* **2020**, *101*, 032308. https://doi.org/10.1103/PhysRevA.101.032308.
5. McClean, J.R.; Boixo, S.; Smelyanskiy, V.N.; Babbush, R.; Neven, H. Barren plateaus in quantum neural network training landscapes. *Nature Communications* **2018**, *9*, 4812. https://doi.org/10.1038/s41467-018-07090-4.
6. Larocca, M.; Thanasilp, S.; Wang, S.; Sharma, K.; Biamonte, J.; Coles, P.J.; Cincio, L.; McClean, J.R.; Holmes, Z.; Cerezo, M. Barren plateaus in variational quantum computing. *Nature Reviews Physics* **2025**, *7*, 174–189. arXiv:2405.00781, https://doi.org/10.1038/s42254-025-00813-9.
7. Cerezo, M.; Sone, A.; Volkoff, T.; Cincio, L.; Coles, P.J. Cost function dependent barren plateaus in shallow parametrized quantum circuits. *Nature Communications* **2021**, *12*, 1791. https://doi.org/10.1038/s41467-021-21728-w.
8. Holmes, Z.; Sharma, K.; Cerezo, M.; Coles, P.J. Connecting ansatz expressibility to gradient magnitudes and barren plateaus. *PRX Quantum* **2022**, *3*, 010313. https://doi.org/10.1103/PRXQuantum.3.010313.
9. Grant, E.; Wossnig, L.; Ostaszewski, M.; Benedetti, M. An initialization strategy for addressing barren plateaus in parametrized quantum circuits. *Quantum* **2019**, *3*, 214. https://doi.org/10.22331/q-2019-12-09-214.
10. Wang, S.; Fontana, E.; Cerezo, M.; Sharma, K.; Sone, A.; Cincio, L.; Coles, P.J. Noise-induced barren plateaus in variational quantum algorithms. *Nature Communications* **2021**, *12*, 6961. https://doi.org/10.1038/s41467-021-27045-6.
11. Schuld, M.; Killoran, N. Is quantum advantage the right goal for quantum machine learning? *PRX Quantum* **2022**, *3*, 030101. https://doi.org/10.1103/PRXQuantum.3.030101.

12. Bowles, J.; Ahmed, S.; Schuld, M. Better than classical? The subtle art of benchmarking quantum machine learning models. *arXiv preprint arXiv:2403.07059* **2024**. https://doi.org/10.48550/arXiv.2403.07059.
13. Kandala, A.; Mezzacapo, A.; Temme, K.; Takita, M.; Brink, M.; Chow, J.M.; Gambetta, J.M. Hardware-efficient variational quantum eigensolver for small molecules and quantum magnets. *Nature* **2017**, *549*, 242–246. https://doi.org/10.1038/nature23879.
14. Pesah, A.; Cerezo, M.; Wang, S.; Volkoff, T.; Sornborger, A.T.; Coles, P.J. Absence of barren plateaus in quantum convolutional neural networks. *Physical Review X* **2021**, *11*, 041011. https://doi.org/10.1103/PhysRevX.11.041011.
15. Mitarai, K.; Negoro, M.; Kitagawa, M.; Fujii, K. Quantum circuit learning. *Physical Review A* **2018**, *98*, 032309. https://doi.org/10.1103/PhysRevA.98.032309.
16. Schuld, M.; Bergholm, V.; Gogolin, C.; Izaac, J.; Killoran, N. Evaluating analytic gradients on quantum hardware. *Physical Review A* **2019**, *99*, 032331. https://doi.org/10.1103/PhysRevA.99.032331.
17. Cerezo, M.; Larocca, M.; García-Martín, D.; et al. Does provable absence of barren plateaus imply classical simulability? *Nature Communications* **2025**, *16*, 7907. arXiv:2312.09121, https://doi.org/10.1038/s41467-025-63099-6.
18. Eshyana Project. Eshyana: An educational quantum computing simulator, 2026. https://quantum.qbithabib.com.
19. Pedregosa, F.; Varoquaux, G.; Gramfort, A.; Michel, V.; Thirion, B.; Grisel, O.; Blondel, M.; Prettenhofer, P.; Weiss, R.; Dubourg, V.; et al. Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research* **2011**, *12*, 2825–2830.
20. Ke, G.; Meng, Q.; Finley, T.; Wang, T.; Chen, W.; Ma, W.; Ye, Q.; Liu, T.Y. LightGBM: A Highly Efficient Gradient Boosting Decision Tree. In Proceedings of the 31st International Conference on Neural Information Processing Systems (NIPS'17), Red Hook, NY, USA, 2017; pp. 3149–3157.